

G3 Applications of differentiation

Name: _____

Class: _____

Date: _____

Time: **469 minutes**

Marks: **390 marks**

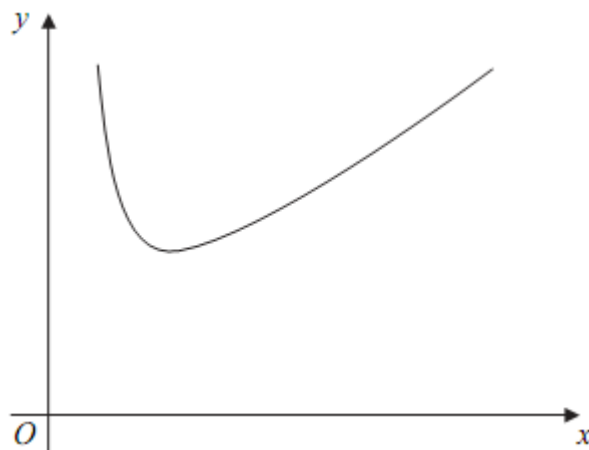
Comments:

EXAM PAPERS PRACTICE

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Q1.

A curve C is defined for $x > 0$ by the equation $y = x + 3 + \frac{8}{x^4}$ and is sketched below.



- (a) Given that $y = x + 3 + \frac{8}{x^4}$, find $\frac{dy}{dx}$. (3)
- (b) Find an equation of the tangent at the point on the curve C where $x = 1$. (3)
- (c) The curve C has a minimum point M . Find the coordinates of M . (4)
- (d) (i) Find $\int \left(x + 3 + \frac{8}{x^4} \right) dx$. (3)
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- (ii) Hence find the area of the region bounded by the curve C , the x -axis and the lines $x = 1$ and $x = 2$. (2)
- (e) The curve C is translated by $\begin{bmatrix} 0 \\ k \end{bmatrix}$ to give the curve $y = f(x)$. Given that the x -axis is a tangent to the curve $y = f(x)$, state the value of the constant k . (1)

(Total 16 marks)

Q2.

- (a) Find $\frac{dy}{dx}$ when $y = (x^3 - 1)^6$. (2)

- (b) A curve has equation $y = x \ln x$.

- (i) Find $\frac{dy}{dx}$. (2)

- (ii) Find an equation of the tangent to the curve $y = x \ln x$ at the point on the curve where $x = e$.

(3)

(Total 7 marks)

Q3.

A curve is defined by the parametric equations

$$x = 3e^t, \quad y = e^{2t} - e^{-2t}$$

- (a) (i) Find the gradient of the curve at the point where $t = 0$. (3)

- (ii) Find an equation of the tangent to the curve at the point where $t = 0$. (1)

- (b) Show that the cartesian equation of the curve can be written in the form

$$y = \frac{x^2}{k} - \frac{k}{x^2}$$

where k is an integer.

(2)

(Total 6 marks)

Q4.

The volume, $V \text{ m}^3$, of water in a tank after time t seconds is given by

$$V = \frac{t^3}{4} - 3t + 5$$

- (a) Find $\frac{dV}{dt}$. (2)

- (b) (i) Find the rate of change of volume, in $\text{m}^3 \text{ s}^{-1}$, when $t = 1$.

(2)

- (ii) Hence determine, with a reason, whether the volume is increasing or decreasing when $t = 1$.

(1)

- (c) (i) Find the positive value of t for which V has a stationary value.

(3)

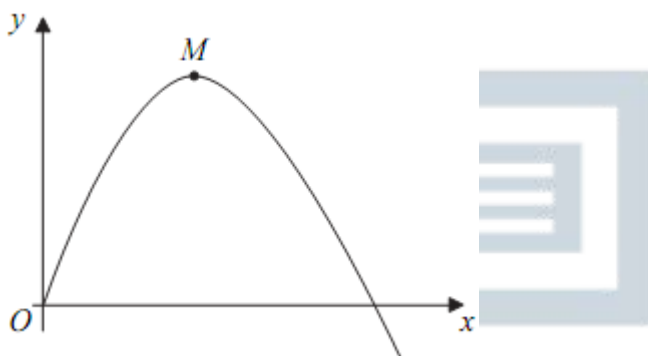
- (ii) Find $\frac{d^2V}{dt^2}$, and hence determine whether this stationary value is a maximum value or a minimum value.

(3)

(Total 11 marks)

Q5.

The diagram shows part of a curve with a maximum point M .



The curve is defined for $x \geq 0$ by the equation

$$y = 6x - 2x^{\frac{3}{2}}$$

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) (i) Hence find the coordinates of the maximum point M .

(3)

- (ii) Write down the equation of the normal to the curve at M .

(1)

- (c) The point $P\left(\frac{9}{4}, \frac{27}{4}\right)$ lies on the curve.

- (i) Find an equation of the normal to the curve at the point P , giving your answer

in the form $ax + by = c$, where a , b and c are positive integers.

(4)

- (ii) The normals to the curve at the points M and P intersect at the point R . Find the coordinates of R .

(2)

(Total 13 marks)

Q6.

- (a) (i) Find $\frac{dy}{dx}$ when $y = xe^{2x}$.

(3)

- (ii) Find an equation of the tangent to the curve $y = xe^{2x}$ at the point $(1, e^2)$

(2)

- (b) Given that $y = \frac{2 \sin 3x}{1 + \cos 3x}$, use the quotient rule to show that

$$\frac{dy}{dx} = \frac{k}{1 + \cos 3x}$$

where k is an integer.

(4)

(Total 9 marks)

Q7.

- (a) A curve is defined by the parametric equations $x = 3 \cos 2\theta$, $y = 2 \cos \theta$.

- (i) Show that $\frac{dy}{dx} = \frac{1}{k \cos \theta}$, where k is an integer.

(4)

- (ii) Find an equation of the normal to the curve at the point where $\theta = \frac{\pi}{3}$.

(4)

- (b) Find the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx$.

(5)

(Total 13 marks)

Q8.

A curve is defined by the equation $2y + e^{2y^2} = x^2 + C$, where C is a constant.

The point $P = \left(1, \frac{1}{e}\right)$ lies on the curve.

- (a) Find the exact value of C .

(1)

- (b) Find an expression for $\frac{dy}{dx}$ in terms of x and y .

(7)

- (c) Verify that $P = \left(1, \frac{1}{e}\right)$ is a stationary point on the curve.

(2)

(Total 10 marks)

Q9.

The depth of water, y metres, in a tank after time t hours is given by

$$y = \frac{1}{8}t^4 - 2t^2 + 4t, \quad 0 \leq t \leq 4$$

- (a) Find:

(i) $\frac{dy}{dt}$,

(3)

(ii) $\frac{d^2y}{dt^2}$.

(2)

- (b) Verify that y has a stationary value when $t = 2$ and determine whether it is a maximum value or a minimum value.

(4)

- (c) (i) Find the rate of change of the depth of water, in metres per hour, when $t = 1$.

(2)

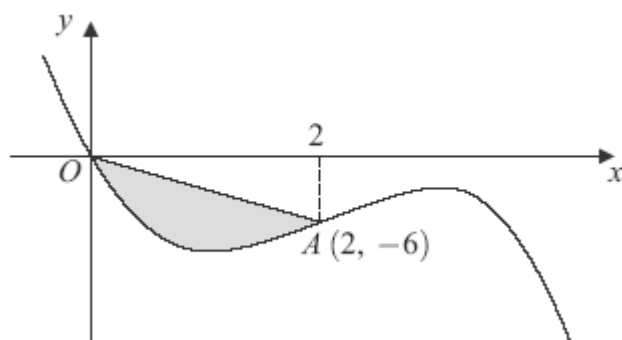
- (ii) Hence determine, with a reason, whether the depth of water is increasing or decreasing when $t = 1$.

(1)

(Total 12 marks)

Q10.

The curve with equation $y = 12x^2 - 19x - 2x^3$ is sketched below.



The curve crosses the x -axis at the origin O , and the point $A(2, -6)$ lies on the curve.

- (a) (i) Find the gradient of the curve with equation $y = 12x^2 - 19x - 2x^3$ at the point A . (4)
- (ii) Hence find the equation of the normal to the curve at the point A , giving your answer in the form $x + py + q = 0$, where p and q are integers. (3)
- (b) (i) Find the value of $\int_0^2 (12x^2 - 19x - 2x^3) dx$. (5)
- (ii) Hence determine the area of the shaded region bounded by the curve and the line OA . (3)
- (Total 15 marks)**

Q11.

A curve has equation $y = \frac{1}{x^3} + 48x$.

- (a) Find $\frac{dy}{dx}$. (3)
- (b) Hence find the equation of each of the two tangents to the curve that are parallel to the x -axis. (4)
- (c) Find an equation of the normal to the curve at the point $(1, 49)$. (3)
- (Total 10 marks)**

Q12.

A curve has equation $y = e^{-4x}(x^2 + 2x - 2)$.

- (a) Show that $\frac{dy}{dx} = 2e^{-4x}(5 - 3x - 2x^2)$. (3)
- (b) Find the exact values of the coordinates of the stationary points of the curve. (5)
- (Total 8 marks)

Q13.

A curve is defined by the equation

$$x^2 + xy = e^y$$

Find the gradient at the point $(-1, 0)$ on this curve.

(Total 5 marks)

Q14.

- (a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$. (2)

- (ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$. (2)

- (b) A curve has parametric equations

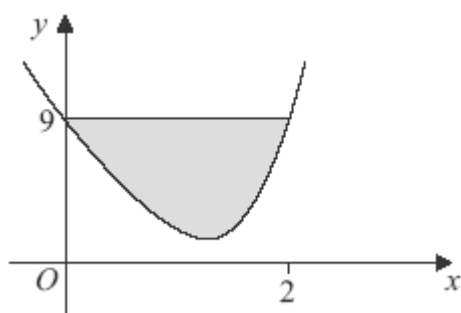
$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

- (i) Find $\frac{dy}{dx}$ in terms of θ . (3)

- (ii) At the point P on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point P . (3)
- (Total 10 marks)

Q15.

The curve with equation $y = x^4 - 8x + 9$ is sketched below.

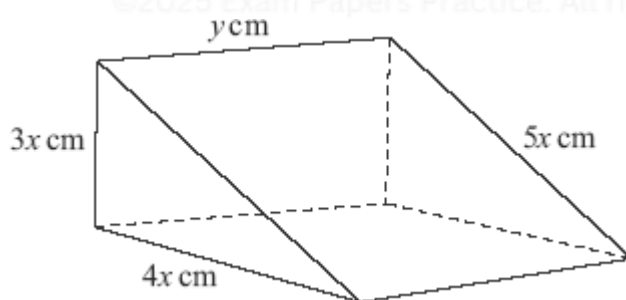


The point (2, 9) lies on the curve.

- (a) (i) Find $\int_0^2 (x^4 - 8x + 9) dx$. (5)
- (ii) Hence find the area of the shaded region bounded by the curve and the line $y = 9$. (2)
- (b) The point $A(1, 2)$ lies on the curve with equation $y = x^4 - 8x + 9$.
- (i) Find the gradient of the curve at the point A . (4)
- (ii) Hence find an equation of the tangent to the curve at the point . (1)
- (Total 12 marks)

Q16.

The diagram shows a block of wood in the shape of a prism with triangular cross-section. The end faces are right-angled triangles with sides of lengths $3x$ cm, $4x$ cm and $5x$ cm, and the length of the prism is y cm, as shown in the diagram.



The total surface area of the five faces is 144 cm^2 .

- (a) (i) Show that $xy + x^2 = 12$. (3)
- (ii) Hence show that the volume of the block, $V \text{ cm}^3$, is given by

$$V = 72x - 6x^3$$

(2)

(b) (i) Find $\frac{dV}{dx}$.

(2)

(ii) Show that V has a stationary value when $x = 2$.

(2)

(c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 2$.

(2)

(Total 11 marks)

Q17.

A curve C has the equation

$$y = \frac{x^3 + \sqrt{x}}{x}, x > 0$$

(a) Express $\frac{x^3 + \sqrt{x}}{x}$ in the form $x^p + x^q$.

(3)

(b) (i) Hence find $\frac{dy}{dx}$.

(2)

(ii) Find an equation of the normal to the curve C at the point on the curve where $x = 1$.

(4)

(c) (i) Find $\frac{d^2y}{dx^2}$.

(2)

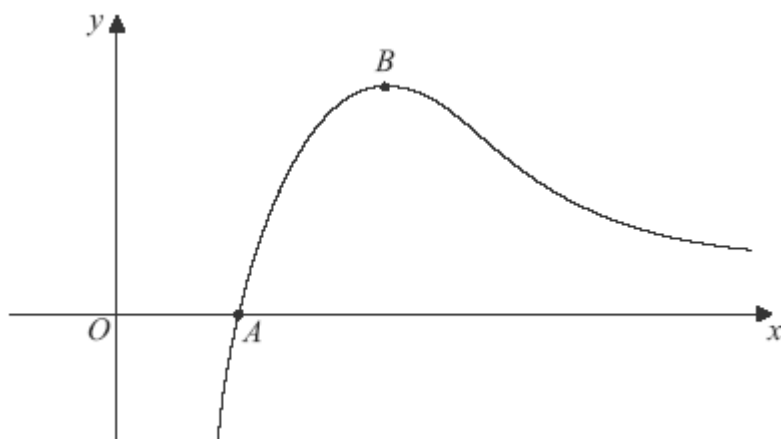
(ii) Hence deduce that the curve C has no maximum points.

(2)

(Total 13 marks)

Q18.

The diagram shows the curve $y = \frac{\ln x}{x}$.



The curve crosses the x -axis at A and has a stationary point at B .

- (a) State the coordinates of A . (1)
 - (b) Find the coordinates of the stationary point, B , of the curve, giving your answer in an exact form. (5)
 - (c) Find the exact value of the gradient of the normal to the curve at the point where $x = e^3$. (3)
- (Total 9 marks)**

Q19.

A curve is defined by the parametric equations

$$x = 1 - 3t, \quad y = 1 + 2t^3$$

- (a) Find $\frac{dy}{dx}$ in terms of t . (3)
 - (b) Find an equation of the normal to the curve at the point where $t = 1$. (4)
 - (c) Find a cartesian equation of the curve. (2)
- (Total 9 marks)**

Q20.

A curve has equation $x^3y + \cos(\pi y) = 7$.

- (a) Find the exact value of the x -coordinate at the point on the curve where $y = 1$. (2)
- (b) Find the gradient of the curve at the point where $y = 1$. (5)
- (Total 7 marks)

Q21.

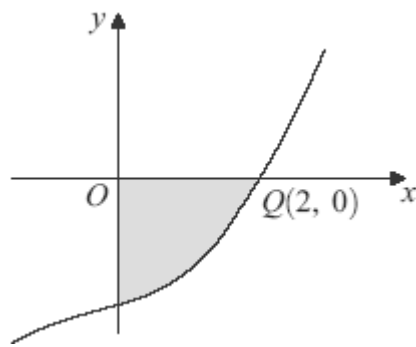
A model car moves so that its distance, x centimetres, from a fixed point O after time t seconds is given by

$$x = \frac{1}{2}t^4 - 20t^2 + 66t, \quad 0 \leq t \leq 4$$

- (a) Find:
- (i) $\frac{dx}{dt}$; (3)
- (ii) $\frac{d^2x}{dt^2}$. (2)
- (b) Verify that x has a stationary value when $t = 3$, and determine whether this stationary value is a maximum value or a minimum value. (4)
- (c) Find the rate of change of x with respect to t when $t = 1$. (2)
- (d) Determine whether the distance of the car from O is increasing or decreasing at the instant when $t = 2$. (2)
- (Total 13 marks)

Q22.

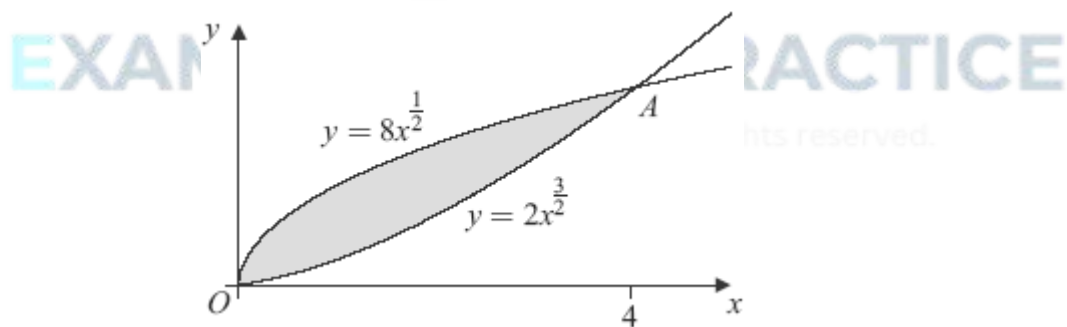
- (a) The polynomial $p(x)$ is given by $p(x) = x^3 + x - 10$.
- (i) Use the Factor Theorem to show that $x - 2$ is a factor of $p(x)$. (2)
- (ii) Express $p(x)$ in the form $(x - 2)(x^2 + ax + b)$, where a and b are constants. (2)
- (b) The curve C with equation $y = x^3 + x - 10$, sketched below, crosses the x -axis at the point $Q(2, 0)$.



- (i) Find the gradient of the curve C at the point Q . (4)
 - (ii) Hence find an equation of the tangent to the curve C at the point Q . (2)
 - (iii) Find $\int (x^3 + x - 10) dx$. (3)
 - (iv) Hence find the area of the shaded region bounded by the curve C and the coordinate axes. (2)
- (Total 15 marks)

Q23.

The diagram shows a sketch of the curves with equations $y = 2x^{\frac{3}{2}}$ and $y = 8x^{\frac{1}{2}}$.



The curves intersect at the origin and at the point A , where $x = 4$.

- (a) (i) For the curve $y = 2x^{\frac{3}{2}}$, find the value of $\frac{dy}{dx}$ when $x = 4$. (2)
- (ii) Find an equation of the normal to the curve $y = 2x^{\frac{3}{2}}$ at the point A . (4)

- (b) (i) Find $\int 8x^{\frac{1}{2}} dx$. (2)
- (ii) Find the area of the shaded region bounded by the two curves. (4)
- (c) Describe a single geometrical transformation that maps the graph of $y = 2x^{\frac{3}{2}}$ onto the graph of $y = 2(x+3)^{\frac{3}{2}}$. (2)
- (Total 14 marks)**

Q24.

- (a) By using the binomial expansion, or otherwise, express $(1 + 2x)^4$ in the form $1 + ax + bx^2 + cx^3 + 16x^4$ where a , b and c are integers. (4)
- (b) Hence show that $(1 + 2x)^4 + (1 - 2x)^4 = 2 + 48x^2 + 32x^4$. (3)
- (c) Hence show that the curve with equation $y = (1 + 2x)^4 + (1 - 2x)^4$ has just one stationary point and state its coordinates. (4)
- (Total 11 marks)**

Q25.

A curve has equation $y = e^{2x}(x^2 - 4x - 2)$.

- (a) Find the value of the x -coordinate of each of the stationary points of the curve. (6)
- (b) (i) Find $\frac{d^2y}{dx^2}$. (2)
- (ii) Determine the nature of each of the stationary points of the curve. (2)
- (Total 10 marks)**

Q26.

A curve is defined by the equation $x^2y + y^3 = 2x + 1$.

- (a) Find the gradient of the curve at the point (2, 1).

(6)

- (b) Show that the x -coordinate of any stationary point on this curve satisfies the equation

$$\frac{1}{x^3} = x + 1$$

(4)

(Total 10 marks)

Q27.

The curve with equation $y = x^5 + 20x^2 - 8$ passes through the point P , where $x = -2$.

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) Verify that the point P is a stationary point of the curve.

(2)

- (c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P .

(3)

- (ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point.

(1)

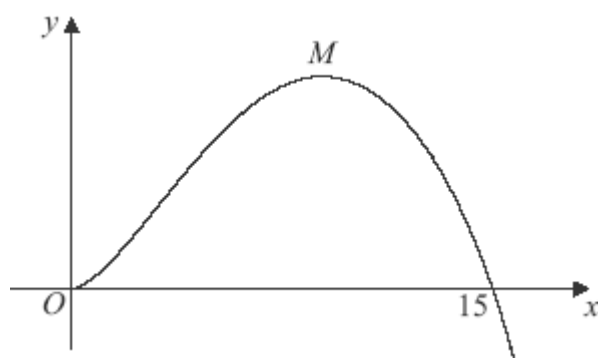
- (d) Find an equation of the tangent to the curve at the point where $x = 1$.

(4)

(Total 13 marks)

Q28.

The diagram shows part of a curve with a maximum point M .



The equation of the curve is

$$y = 15x^{\frac{3}{2}} - x^{\frac{5}{2}}$$

- (a) Find $\frac{dy}{dx}$. (3)
- (b) Hence find the coordinates of the maximum point M . (4)
- (c) The point $P(1, 14)$ lies on the curve. Show that the equation of the tangent to the curve at P is $y = 20x - 6$. (3)
- (d) The tangents to the curve at the points P and M intersect at the point R . Find the length of RM . (3)
- (Total 13 marks)

Q29.

- (a) The curve with equation

$$y = \frac{\cos x}{2x+1}, \quad x > -\frac{1}{2}$$

intersects the line $y = \frac{1}{2}$ at the point where $x = \alpha$.

- (i) Show that α lies between 0 and $\frac{\pi}{2}$. (2)
- (ii) Show that the equation $\frac{\cos x}{2x+1} = \frac{1}{2}$ can be rearranged into the form

$$x = \cos x - \frac{1}{2}$$

(1)

- (iii) Use the iteration $x_{n+1} = \cos x_n - \frac{1}{2}$ with $x_1 = 0$ to find x_3 , giving your answer to three decimal places.

(2)

- (b) (i) Given that $y = \frac{\cos x}{2x+1}$, use the quotient rule to find an expression for $\frac{dy}{dx}$.

(3)

- (ii) Hence find the gradient of the normal to the curve $y = \frac{\cos x}{2x+1}$ at the point on the curve where $x = 0$.

(2)

(Total 10 marks)

Q30.

A curve is defined by the parametric equations

$$x = \frac{1}{t}, \quad y = t + \frac{1}{2t}$$

- (a) Find $\frac{dy}{dx}$ in terms of t .

(4)

- (b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

- (c) Show that the cartesian equation of the curve can be written in the form

$$x^2 - 2xy + k = 0$$

where k is an integer.

(3)

(Total 11 marks)

Q31.

A curve is defined by the equation $4x^2 + y^2 = 4 + 3xy$.

Find the gradient at the point (1, 3) on this curve.

(Total 5 marks)

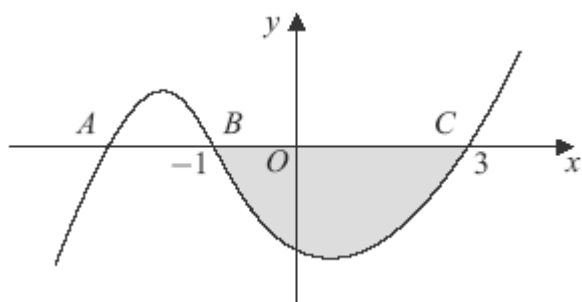
Q32.

The curve with equation $y = x^4 - 32x + 5$ has a single stationary point, M .

- (a) Find $\frac{dy}{dx}$. (3)
- (b) Hence find the x -coordinate of M . (3)
- (c) (i) Find $\frac{d^2y}{dx^2}$. (1)
- (ii) Hence, or otherwise, determine whether M is a maximum or a minimum point. (2)
- (d) Determine whether the curve is increasing or decreasing at the point on the curve where $x = 0$. (2)
- (Total 11 marks)

Q33.

- (a) The polynomial $p(x)$ is given by $p(x) = x^3 - 7x - 6$.
- (i) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$. (2)
- (ii) Express $p(x) = x^3 - 7x - 6$ as the product of three linear factors. (3)
- (b) The curve with equation $y = x^3 - 7x - 6$ is sketched below.



The curve cuts the x -axis at the point A and the points $B(-1, 0)$ and $C(3, 0)$.

- (i) State the coordinates of the point A . (1)

(ii) Find $\int_{-1}^3 (x^3 - 7x - 6) dx$. (5)

(iii) Hence find the area of the shaded region bounded by the curve $y = x^3 - 7x - 6$ and the x -axis between B and C . (1)

(iv) Find the gradient of the curve $y = x^3 - 7x - 6$ at the point B . (3)

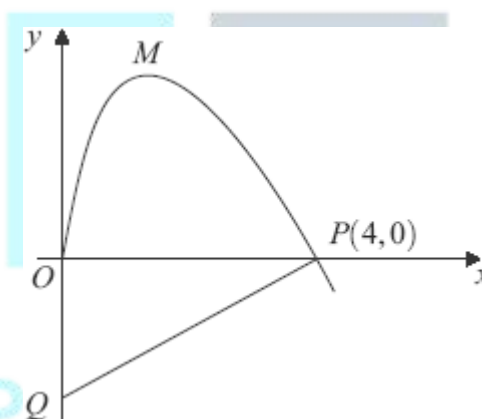
(v) Hence find an equation of the normal to the curve at the point B . (3)

(Total 18 marks)

Q34.

A curve, drawn from the origin O , crosses the x -axis at the point $P(4, 0)$.

The normal to the curve at P meets the y -axis at the point Q , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = 4x^{\frac{1}{2}} - x^{\frac{3}{2}}$$

(a) (i) Find $\frac{dy}{dx}$. (3)

(ii) Show that the gradient of the curve at $P(4, 0)$ is -2 . (2)

(iii) Find an equation of the normal to the curve at $P(4, 0)$. (3)

(iv) Find the y -coordinate of Q and hence find the area of triangle OPQ .

(3)

- (v) The curve has a maximum point M . Find the x -coordinate of M .

(3)

(b) (i) Find $\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx$.

(3)

- (ii) Find the total area of the region bounded by the curve and the lines PQ and QO .

(3)

(Total 20 marks)

Q35.

A curve is defined by the parametric equations $x = 2t + \frac{1}{t^2}$, $y = 2t - \frac{1}{t^2}$.

- (a) At the point P on the curve, $t = \frac{1}{2}$.

- (i) Find the coordinates of P .

(2)

- (ii) Find an equation of the tangent to the curve at P .

(5)

- (b) Show that the cartesian equation of the curve can be written as

$$(x - y)(x + y)^2 = k$$

where k is an integer.

(3)

(Total 10 marks)